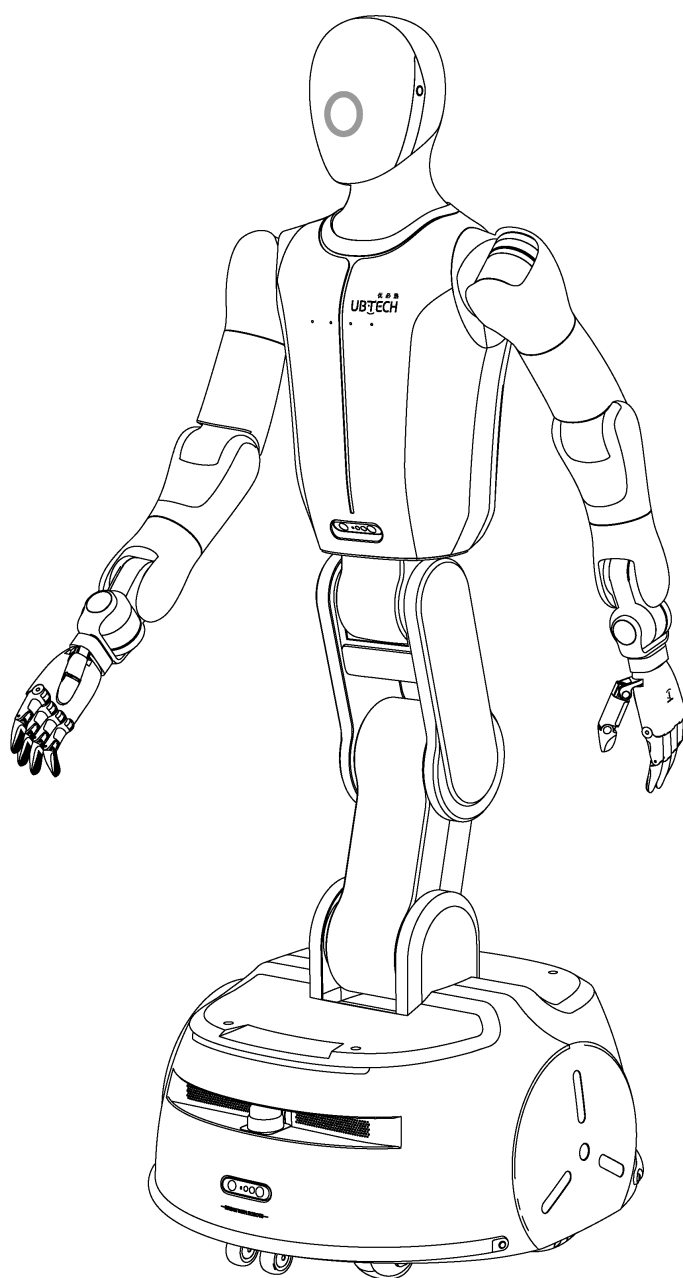


Product Manual

CruZr S2



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1.Disclaimer

- This manual is only an operation and maintenance guide document for the CRUZR S2 Wheeled Humanoid Robot (hereinafter referred to as the "Robot") of SHENZHEN UBTECH Robotics Corp., Ltd. (hereinafter referred to as "UBTECH"). All contents are compiled based on the hardware configuration, software version and technical parameters of the Robot at the time of delivery. If the Robot undergoes subsequent hardware modifications, software upgrades or parameter adjustments, the updated manual document officially released by UBTECH shall prevail.
- The Robot is applicable to designated tasks in industrial or commercial scenarios (e.g., industrial scenarios: box handling, quality inspection, SPS sorting, etc.; commercial scenarios: navigation, reception, etc.). It is strictly prohibited to use the Robot in non-industrial and non-commercial scenarios such as medical and military applications. UBTECH shall not bear any responsibility for any personal injury or property loss caused by improper scenario use.
- The Robot must be operated in strict accordance with this manual and the official training requirements of UBTECH. UBTECH shall not be liable for compensation for equipment failures, production accidents or personal injuries caused by non-compliant operations (such as failure to maintain a safe distance, unauthorized modification of core parameters, failure to perform maintenance on time, etc.).
- The Robot may experience performance fluctuations or functional abnormalities in complex industrial environments (such as strong electromagnetic interference, extreme temperatures, severe vibrations, etc.). UBTECH only provides relevant maintenance services in accordance with the official after-sales policy of the Robot for the normal operation of the equipment under compliant operating environment requirements. The user shall bear the cost of repair or replacement for equipment problems caused by non-compliant operating environments.

2. Safety Instructions

2.1 Personnel Safety

- When the Robot is in operation, it is forbidden to stand, stay or place irrelevant items within its activity range (especially the operating areas of the dual arms, folding mechanism and dexterous hands) to avoid collision injuries caused by mechanical movement; during operation, a safe distance of at least 1.5 meters must be maintained from the Robot.
- It is strictly prohibited for non-professional personnel to disassemble core components of the Robot such as the outer shell, battery, sensors or servo drivers. Maintenance or servicing must be performed by UBTECH-certified engineers to prevent safety accidents caused by electric shock or component damage.
- When the Robot performs high-risk tasks, it must be equipped with exclusive protective devices (such as anti-corrosion shells, torque overload protection). Operators must wear protective equipment such as chemical protective clothing and safety helmets, and monitor the equipment operation status in real time.

2.2 Equipment Safety

- The maximum load capacity of the Robot's dual arms is 15kg. Overload operation is strictly prohibited to avoid damage to the joint motors due to overload or equipment damage caused by falling heavy objects.
- During charging, only the charger or smart charging pile independently developed by UBTECH shall be used. The use of third-party charging equipment is prohibited to prevent fire hazards caused by battery overcharging or short circuit.
- The emergency stop switch (marked at a prominent position on the Robot body) is a mandatory shutdown device for emergency situations. It shall only be used when the Robot is out of control, there is a collision risk or personnel are in danger. It is forbidden to trigger it randomly during daily operations.

2.3 Environmental Safety

- The working area of the Robot must have a flat ground without water accumulation or oil stains to prevent the Robot from slipping during movement; at the same time, obstacles such as cables and tools in the area must be cleaned up to prevent the Robot's wheels from being entangled and affecting movement efficiency.
- The use of the Robot in flammable and explosive environments (such as warehouses storing alcohol, gasoline) or environments with strong electromagnetic interference (such as near large transformers, high-frequency welding equipment) is prohibited to prevent fire, explosion or equipment out of control caused by sensor signal disorder.

3. Packing List

- ✧ Robot Body * 1
- ✧ Power Charger * 1
- ✧ Remote Control * 1
- ✧ Product Manual * 1
- ✧ Certificate of Conformity * 1
- ✧ Warranty Card * 1
- ✧ Factory Inspection Report * 1

4. Product Overview

4.1 Product Introduction

Cruzr S2 is UBTECH's new-generation general-purpose wheeled humanoid robot, developed based on full-stack humanoid robot technology and integrated with key

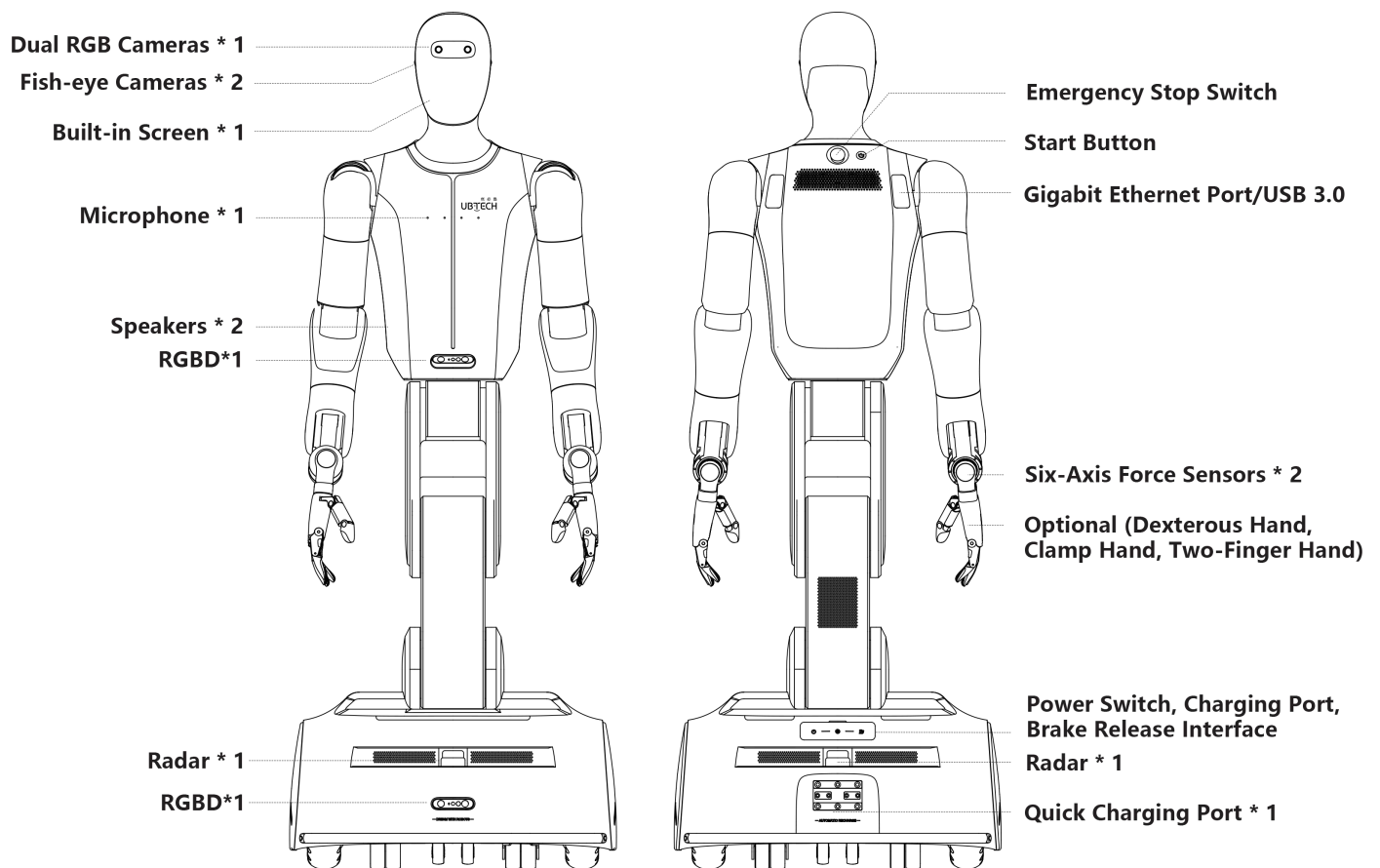
embodied intelligence technologies such as multi-modal planning large models. It features human-like form, high load capacity, autonomous navigation and multi-scenario adaptability.

The entire Robot adopts a full-joint hollow internal wiring design, equipped with joint encoders with dual encoders and a local air-cooling heat dissipation system, which effectively improves operation accuracy and operation stability. The main structure of the Robot body is made of high-strength materials such as aerospace aluminum alloy, with a total weight of 188kg. The entire machine (excluding dexterous hands) has 22 joint degrees of freedom, and each arm has 7 degrees of freedom, endowing it with excellent operational flexibility.

Equipped with a high-performance GPU and depth camera, it supports Wi-Fi 6 and Bluetooth 5.2 wireless communication to achieve efficient data interaction. Its end effector can be optionally configured and quickly replaced among dexterous hands, clamp hands and two-finger grippers to meet the diverse needs of customers.

In commercial scenarios, Cruzr S2 aims to achieve "unlimited possibilities of intelligent services", covering scenarios such as reception and navigation in exhibition halls, medical drug transportation in pharmacies, and sorting and shopping guidance in retail supermarkets. In industrial scenarios, it focuses on "alleviating the labor shortage in the manufacturing industry", which can adapt to unstructured environments, perform high-repeatability/high-risk tasks, and collaborate with unmanned logistics equipment and intelligent manufacturing systems to promote the flexible and unmanned upgrading of the industry.

4.2 Introduction to Key Components



(Figure 3-2-1)

4.3 Specifications

Parameter Category	Specific Parameters
Height	176cm
Arm Span	177cm (excluding dexterous hands)
Weight	188kg (excluding dexterous hands, full counterweight)
Head Degrees of Freedom	2
Degrees of Freedom per Arm (excluding dexterous hands)	7
Waist Degrees of Freedom	1
Folding Mechanism Degrees of Freedom	3
Wheeled Chassis Degrees of Freedom	2
Load Capacity	Maximum load of dual arms: 15kg
Obstacle Crossing Height	$\leq 20\text{mm}$
Gap Crossing Width	$< 50\text{mm}$
Climbing Angle	$\leq 5^\circ$
Computing Power Configuration	X86 + NVIDIA Jetson Orin (275TOPS)
Sensor Configuration	Depth camera, six-axis force sensor, IMU, fisheye camera

Display and Interaction	4-inch circular interactive screen, battery status indicator, microphone array, speaker
Power Supply Method	Lithium battery DC 48V
Battery Charging Voltage/Current	Voltage: DC 54V; Current: 20A
Comprehensive Battery Life	8h
Charging Time	≤ 3 h
Intelligent OTA Upgrade	Available
Operating System	Ubuntu + ROSA 2.0

5. User Instructions

5.1 Operating Environment Requirements

5.1.1 Spatial Environment

1. Working area size: It shall meet the activity range of the Robot. The minimum working space for a single Robot is not less than 4 m² (L2m × W2m). For collaborative operation scenarios, a passage for equipment intersection shall be reserved (width $\geq$ 1.5 meters).
2. Ground conditions: The ground shall be flat, with no water accumulation, oil stains or sharp debris; the ground load-bearing capacity $\geq$ 250kg/m² (to prevent the ground from sinking when the Robot moves).

5.1.2 Temperature, Humidity and Lighting

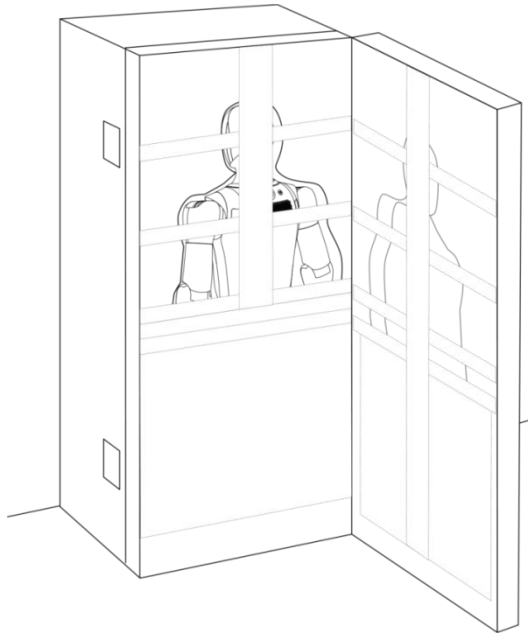
1. Temperature range: 0~40°C. It is forbidden to use the Robot in environments below 0°C or above 40°C. Using the Robot beyond this temperature range will reduce battery performance and affect battery service life.
2. Humidity range: 20%~80% (non-condensing). Dehumidification equipment shall be equipped in humid environments to prevent fogging on sensor lenses or rusting of metal components.
3. Lighting conditions: The illumination intensity of the working area $\geq 200\text{lux}$. Avoid direct strong light (such as direct sunlight on camera lenses) or excessively dim light to ensure the accuracy of visual positioning and recognition.

5.1.3 Electromagnetic and Network Environment

1. Electromagnetic interference: Keep away from strong electromagnetic radiation sources (such as high-frequency welding machines, large transformers) to prevent sensor signal disorder.
2. Network requirements: Gigabit Ethernet is supported, with network latency $\leq 100\text{ms}$ and packet loss rate $\leq 1\%$, to ensure real-time data transmission between the Robot, the cloud and collaborative equipment.

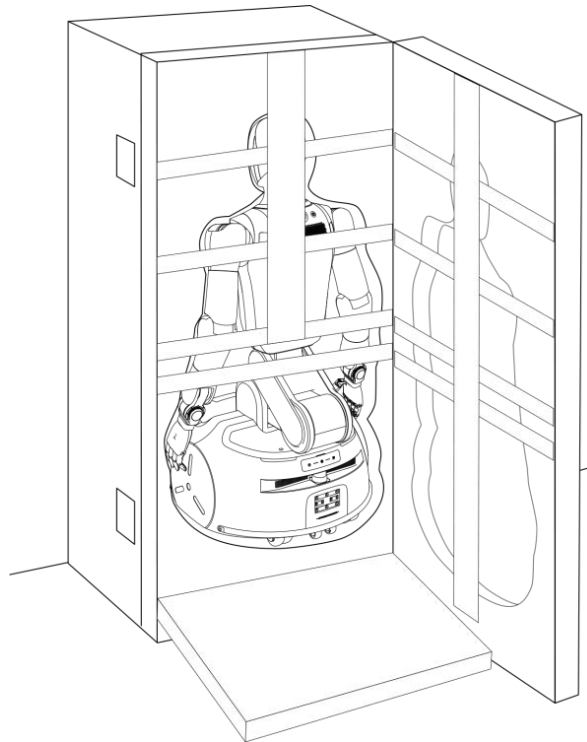
5.2 Unpacking Operation

1. Place the air transport case horizontally and lock the wheels.
2. Open the case door. (See Figure 5-2-1)



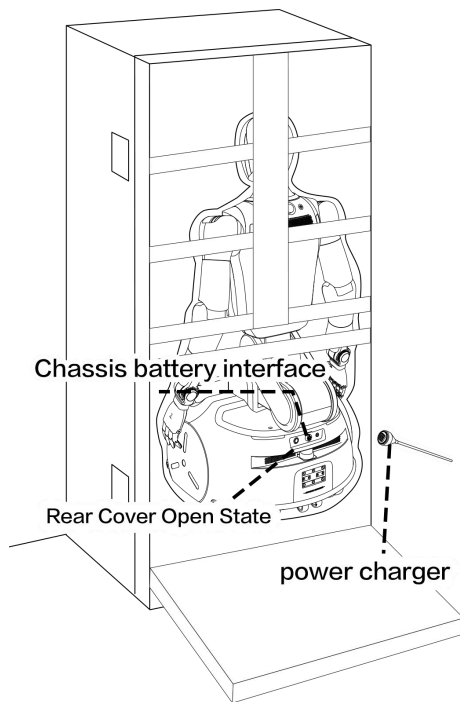
(Figure 5-2-1)

3. Remove the binding straps and lower the upright ramp board. (See Figure 5-2-2)

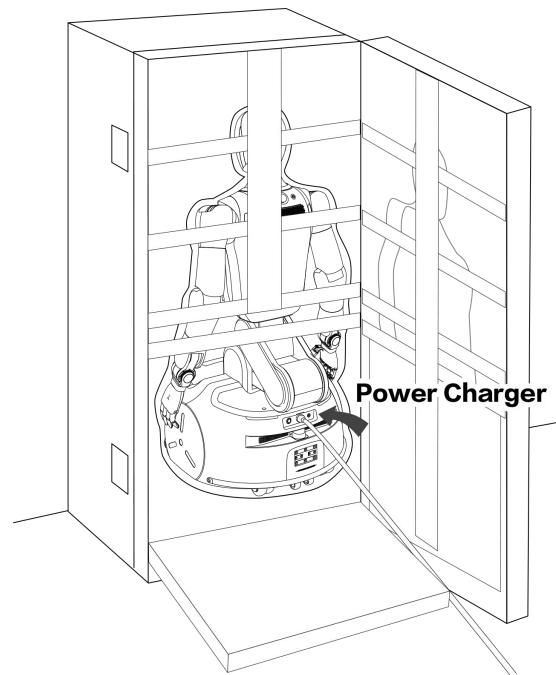


(Figure 5-2-2)

- ✧ **Note:** For long-distance transportation (>3 hours), battery wake-up is required. Wake-up method: Connect the power charger to the chassis battery interface and plug in the power supply. (See Figure 5-2-3) (See Figure 5-2-4)

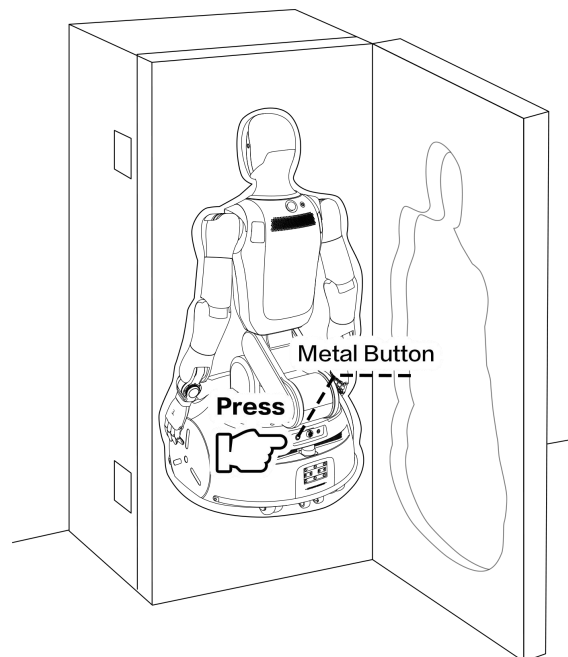


(Figure 5-2-3)



(Figure 5-2-4)

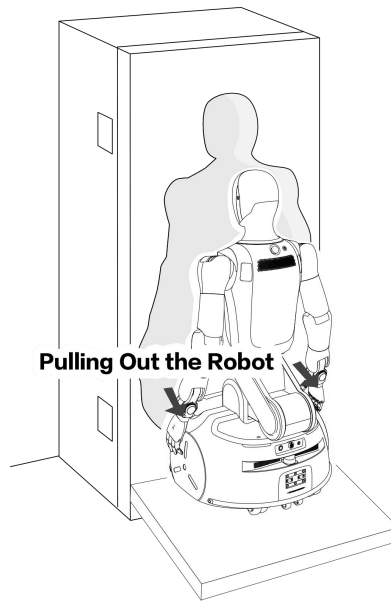
4. Take out the EPE foam.
5. Press the metal button on the chassis to start the chassis and release the chassis brake. (See Figure 5-2-5)



(Figure 5-2-5)

6. Pull out the Robot, take out the remote control and power supply, and close the

case door to complete the unpacking operation. (See Figure 5-2-6)

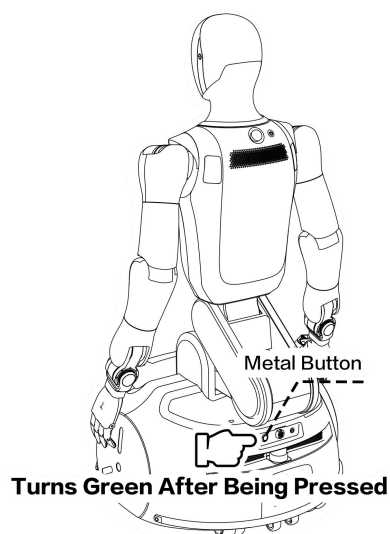


(Figure 5-2-6)

5.3 Power-On, Power-Off and Emergency Stop Operations

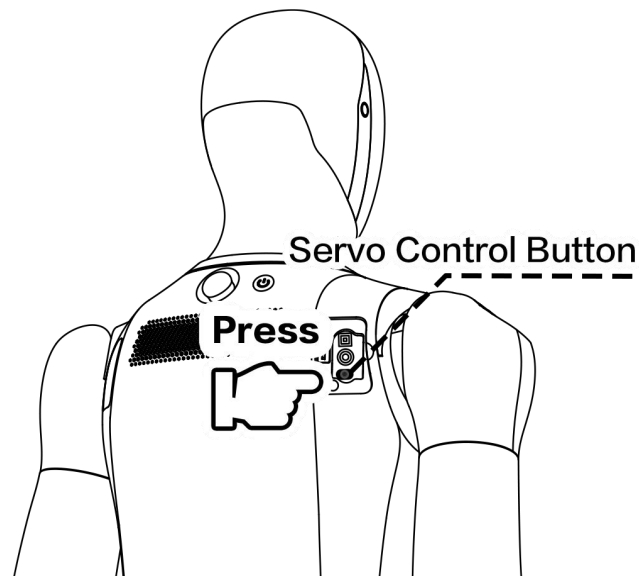
5.3.1 Power-On Operation

1. Before power-on, ensure that the emergency stop button on the back of the Robot is in the released state.
2. Press the metal button on the chassis, and the button will turn green. (See Figure 5-3-1)



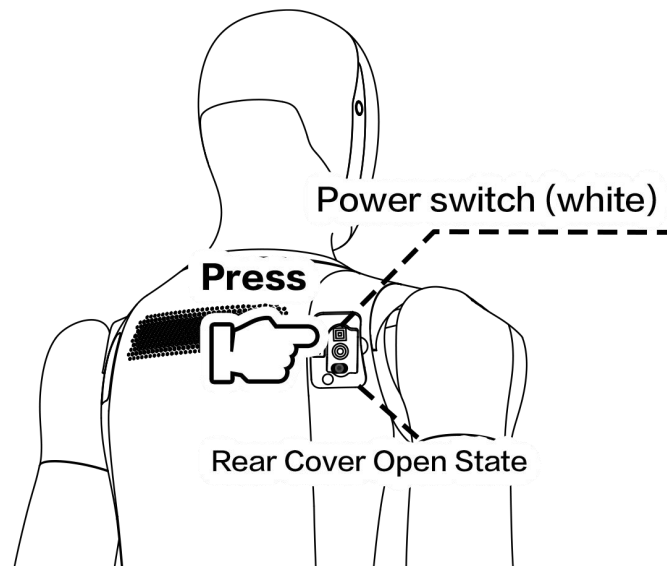
(Figure 5-3-1)

3. Open the back cover of the Robot, press the servo control button, and ensure that the black servo emergency stop button inside the cover is in the popped-up state. (See Figure 5-3-2)



(Figure 5-3-2)

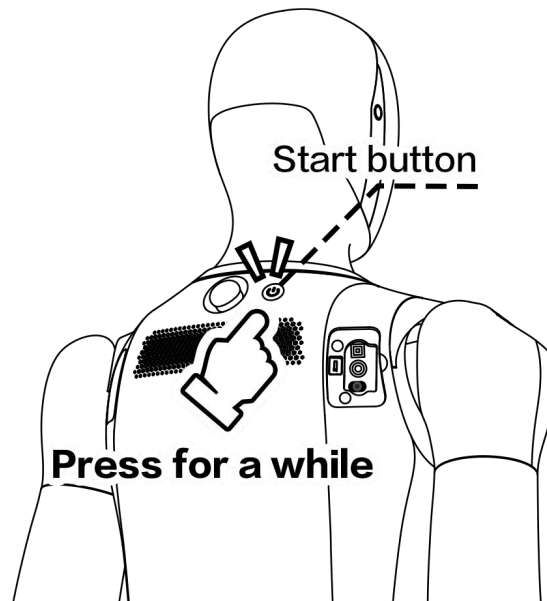
4. Press the power switch on the back of the Robot. (See Figure 5-3-3)



(Figure 5-3-3)

5. Press the start button for a while on the back of the Robot for about 5 seconds.

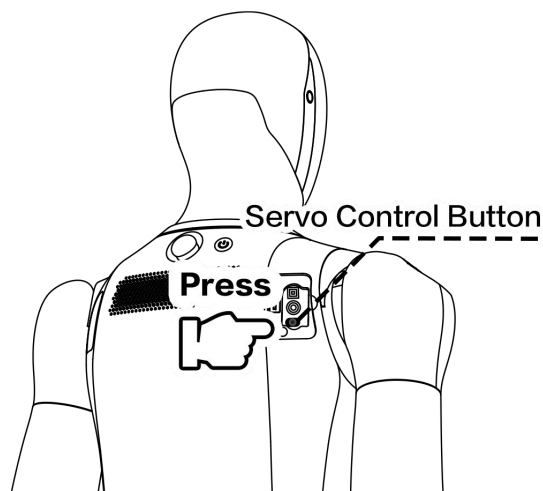
When the display screen on the Robot's head lights up, it indicates that the Robot has started to power on. The start-up time is about 2 minutes. (See Figure 5-3-4)



(Figure 5-3-4)

5.3.2 Power-Off Operation

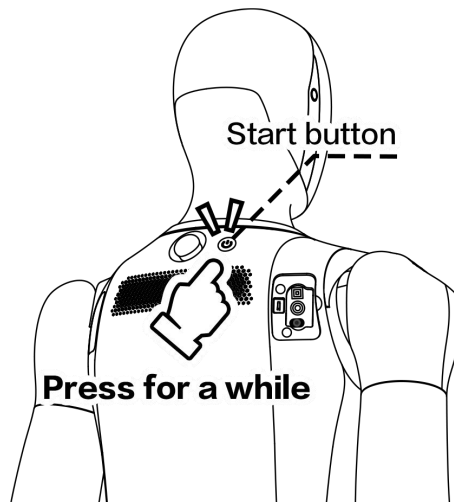
1. Ensure the Robot is in damping mode.
2. Open the back cover of the Robot and press the servo control button. (See Figure 5-3-5)



(Figure 5-3-5)

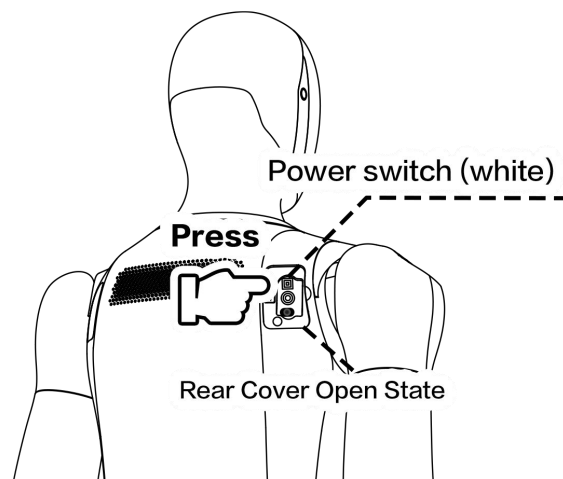
3. Double-click and hold the start button on the back of the Robot, and keep this

state for 5 seconds. The Robot will prompt that it is about to shut down. Keep holding the button, and the Robot will enter the shutdown procedure. Wait until the Robot's screen and lights go out, and the Robot will be in the power-off state. (See Figure 5-3-6)



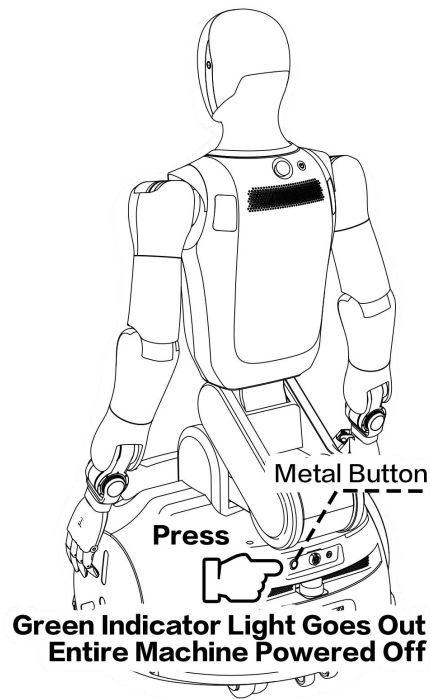
(Figure 5-3-6)

4. Open the back cover of the Robot, press the power switch at the lower rear, and power off the upper body of the Robot. (See Figure 5-3-7)



(Figure 5-3-7)

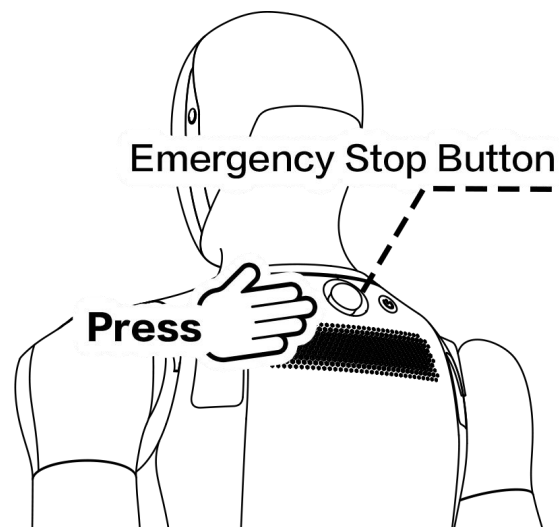
5. Press the chassis button to pop up the metal button, and the green indicator light will go out, indicating that the entire machine is powered off. (See Figure 5-3-8)



(Figure 5-3-8)

5.3.3 Emergency Stop Operation

- In case of an emergency requiring the Robot to stop immediately, press the emergency stop button on the back of the Robot. At this time, all motors of the Robot will be in a locked state (i.e., "statue state") and will no longer accept external command signals. (See Figure 5-3-9)



(Figure 5-3-9)

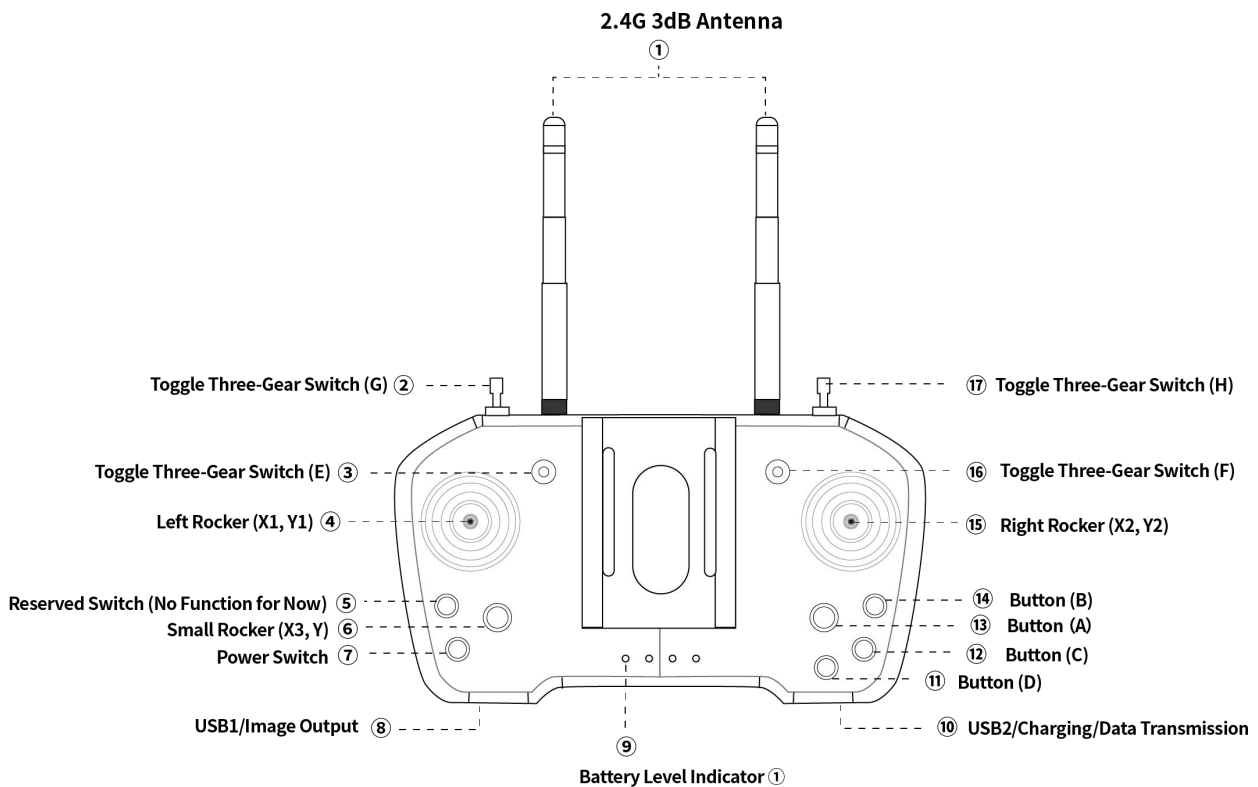
- **Emergency Stop Recovery:** After powering off the Robot, restart the entire machine.

- ✧ **Note:** The emergency stop of the Robot can also be triggered remotely via the remote control. For details, see Remote Control Operation.

5.4 Remote Control Operation

5.4.1 Remote Control Diagram: It is recommended to extend the antenna before use

The remote control is equipped with a built-in integrated rechargeable lithium battery, which is compatible with standard microUSB interface 5V power adapters on the market (such as USB chargers for digital products like mobile phones and cameras) for charging. (See Figure 5-4-1)



(Figure 5-4-1)

5.4.2 Remote Control Btton Description

Function	Buttons	Function	Buttons
Homing	F Down + D	Enter Damping Mode	F Down + C
Forward/Backward Movement	Y1 Push Up/Down	Exit Damping Mode	F Up + C
Folding Mechanism	Y2 Right Rocker (Up: Stand; Down: Sit)	Turning	X2 Right Rocker (Left: Counterclockwise; Right: Clockwise)
Function	Buttons		
Task Flow	E Up (Previous Action), Middle (Current Action) Down (Next Action) G (Right): Execute Action		
Terminate Task Flow	H (Middle) + A (Available during execution) Interrupt Task → Return to Control Mode → Reset Task Flow		
Power-On	Power-On → Homing → Control (Homing is mandatory after abnormal shutdown)		
Power-Off	Homing First, Then Sit Down		
Mandatory Requirement	TTS Broadcast Warning for Power-On/Power-Off		

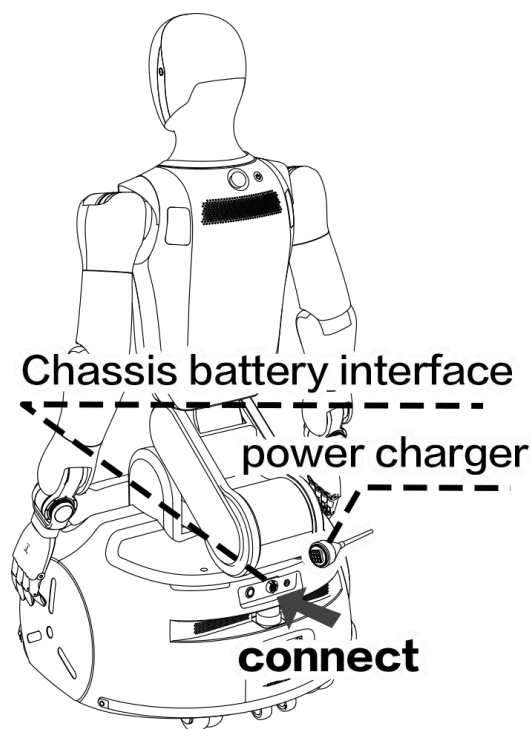
(Figure 5-4-2)

5.4.3 Remote Control Light Signals

Remote Control Battery Level Indicator				
•	•	•	•	75%-100%
•	•	•	◦	50%-75%
•	•	◦	◦	25%-50%
•	◦	◦	◦	0%-25%

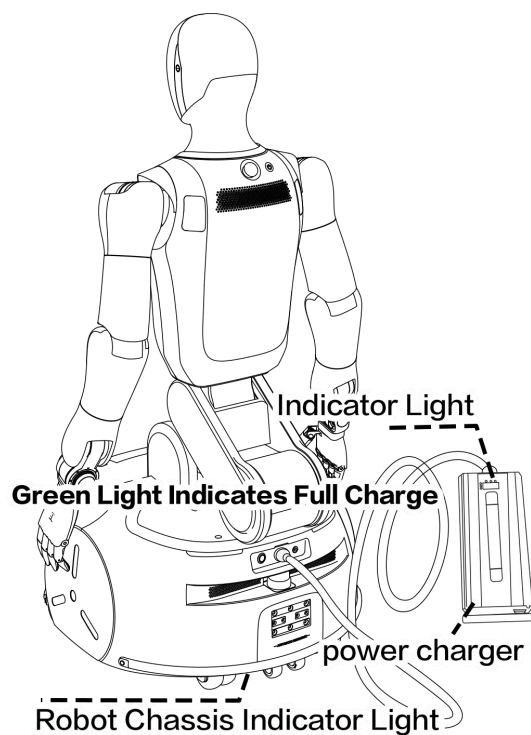
5.5 Charging Operation

1. Connect the charger to the AC power supply (100-240V, 50/60Hz). Before connection, ensure that the external power supply voltage matches the rated input voltage of the charger; otherwise, the charger may be damaged (the rated input voltage of the charger is marked on the charger nameplate).
2. Check whether there are foreign objects in the rotating plug of the charger and the chassis charging port of the Robot.
3. Insert the charger plug into the chassis charging port. (See Figure 5-5-1)



(Figure 5-5-1)

4. If the indicator light of the power charger turns red, it indicates that charging is in progress. If the indicator light of the power charger turns green, it indicates that the battery is fully charged. At this time, charging should be stopped to avoid overcharging the battery. (See Figure 5-5-2)



(Figure 5-5-2)

5. When the indicator light at the bottom of the chassis is solid green, it also indicates that the battery is fully charged. After charging is completed, remove the charger.

Type	Status	Light Type	Voice Broadcast Remarks
Power-On/Power-Off	Powering On	White Flowing Light	No voice
	Powering Off	White Flowing Light	Voice: "Powering off" (broadcast once)
Normal Working Status	Waiting after successful power-on	Solid White	/
	Executing Actions	Blue Flowing Light	/
Abnormal Working	Motion control operation abnormality,	Solid Red	/

Status	Sensor abnormality, Over-limit abnormality, Servo abnormality		
Charging	Charging in Progress	Flashing Green	When charging starts, voice broadcast: "Charging started" (once)
Battery Level Check	/	Short-press the power button, voice broadcast: "Current remaining battery level: X%" (once)	
	Charging Completed	Solid Green	When battery level reaches 100%, voice broadcast: "Charging completed" (once)
Battery Level Alarm	Battery Level Reminder (30%)	Flashing Yellow	Voice broadcast: "Remaining battery level: 30%, please charge as soon as possible" (once)
	Low Battery Level (10%-20%)	Flashing Red	Voice broadcast: "Low battery, please charge as soon as possible" (once every 10 seconds)
	Critical Battery Level ($\leq$ 10%)	Flashing Red	Voice broadcast: "Extremely low battery, please charge immediately" (once every 5 seconds)

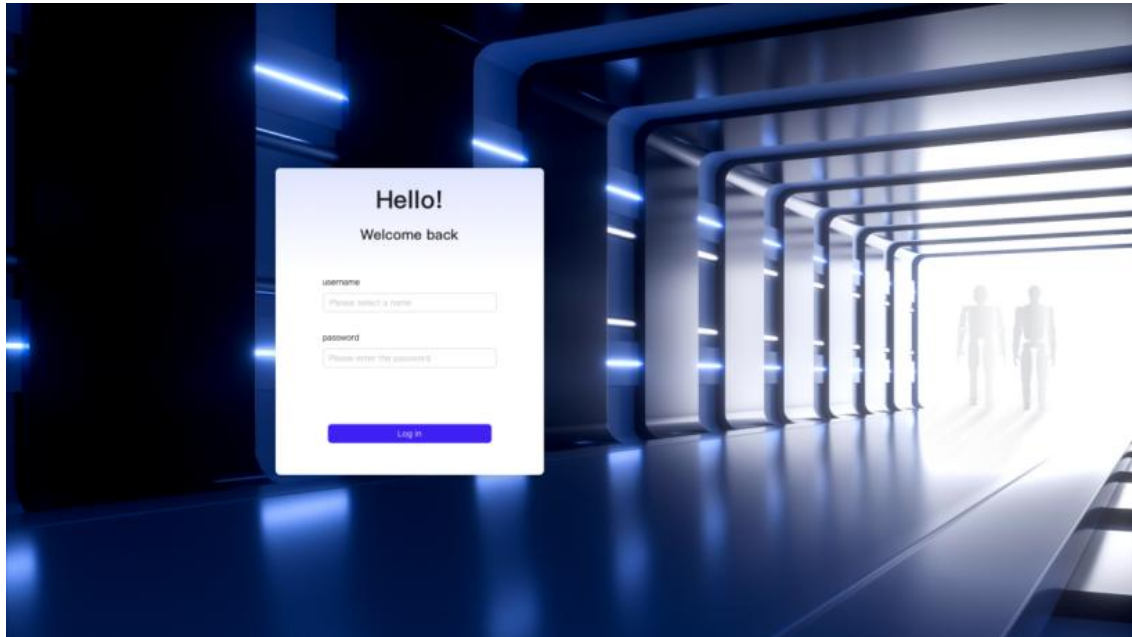
- ✧ **Note:** The battery may have firmware upgrades. The status indicator light shall be subject to the physical product.

5.6 WiFi network configuration operation

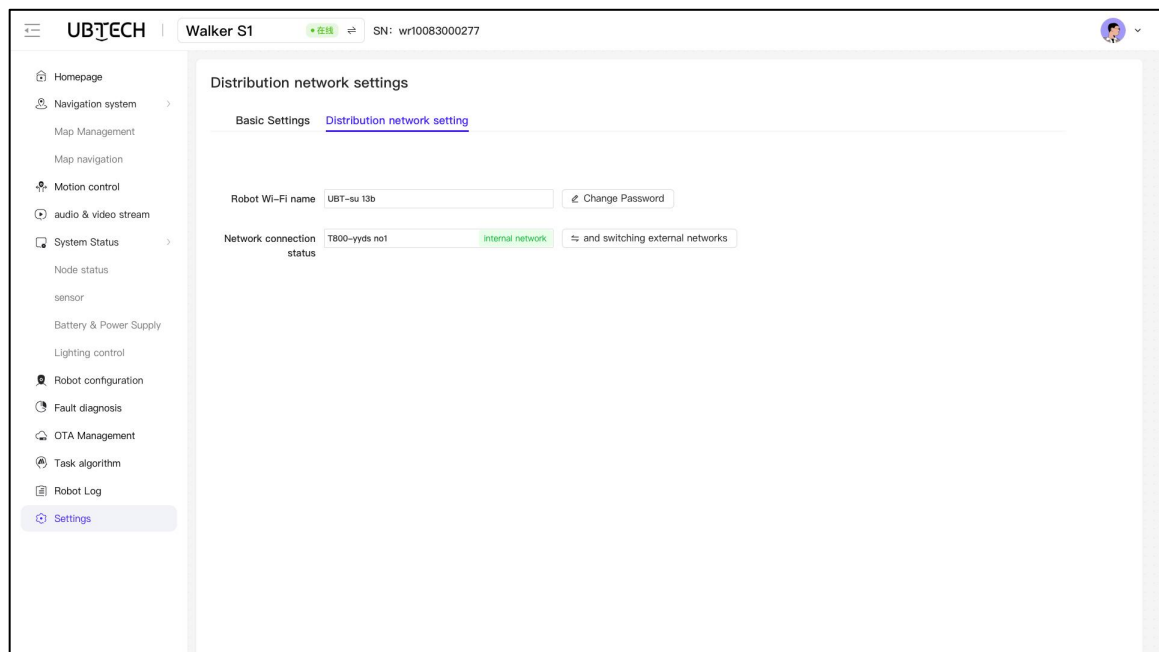
Process of configuring the extranet for the robot on the web client:

Step 1: After turning on the robot, the computer needs to connect to the robot's hotspot WiFi.

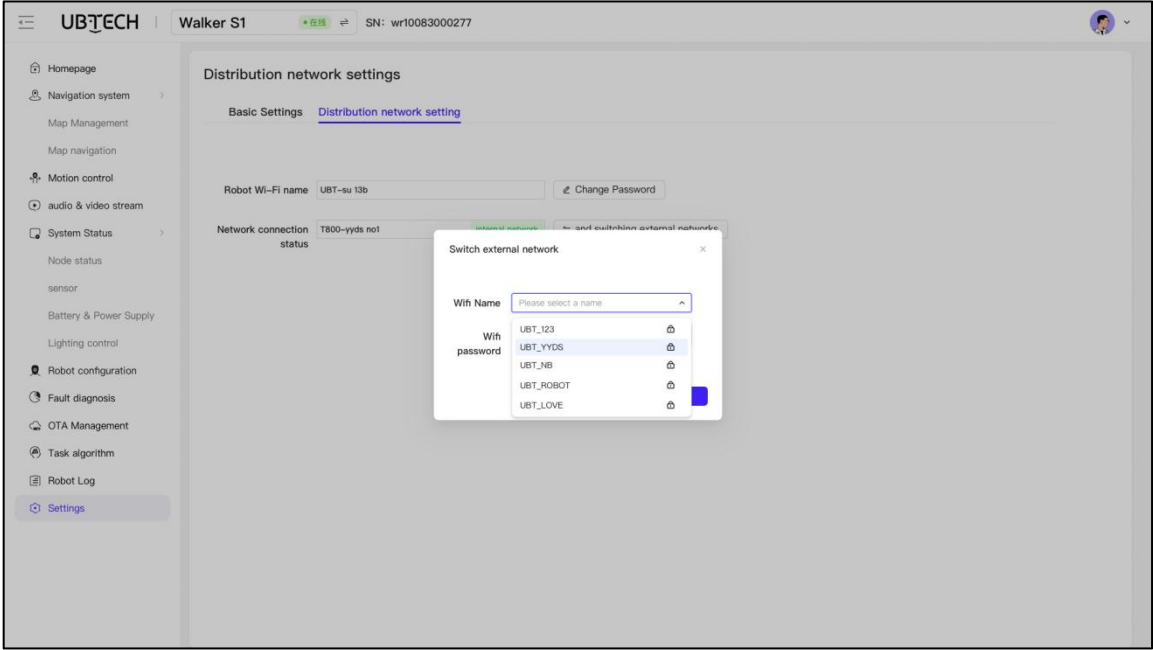
Step 2: After successfully connecting to the robot's hotspot WiFi, open the Web URL on your computer: <http://192.168.11.3>, and enter your username and password.



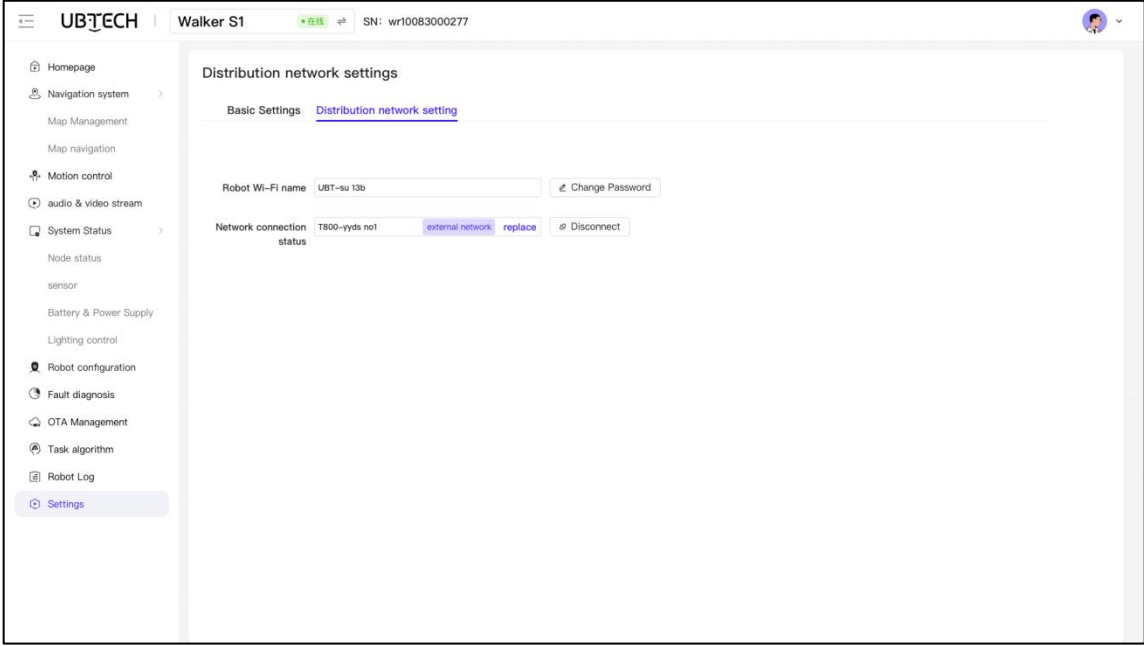
Step 3: On the Web page - Settings, click the "Switch to Extranet" button to configure external WiFi for the robot.



Step 4: Enter the external WiFi to which the robot will connect.



Step 5: Complete the external network configuration operation of the robot.



5.7 Common Faults and Troubleshooting Methods

Possible Problem	Troubleshooting Method
Failure to Start	• Check whether the chassis power button is loose or abnormal; • Check whether the back power-on/power-off button is loose or abnormal; • The battery may be low. Please connect the charger and charge for 10 minutes, then power on; • Check whether the black servo button on the back is pressed; • Check whether the emergency stop switch is pressed; • Check whether the battery is in hibernation. If it is in hibernation, wake-up is required.
Failure to Charge	• Check whether the charger is plugged into the power socket; • Check whether the charger is correctly connected to Cruzr S2; • Check the charger LED indicator status: • Solid Green: Battery is fully charged or the charger is plugged in but not connected to CRUZR S2; • Solid Red: Charger is charging. • If the problem persists after multiple attempts, please contact the UBTECH Technical Support Center.
Overheating of Charger	• If the charger overheats, it may be caused by a fault; please disconnect the charger from the wall socket immediately and contact the UBTECH Technical Support Center.
Six-Axis Force Abnormality	Press the emergency stop button for 10 seconds, then turn it on again.

5.7 Daily Maintenance

5.7.1 Whole-Machine Cleaning

1. Before cleaning the Robot, completely power off the Robot, confirm that the emergency stop button is in the popped-up state, and ensure that the charger is not connected.
2. It is forbidden to use sprays, alcohol-based solutions, water guns or other tools to directly contact the Robot.
3. It is recommended to use a soft dry cloth to wipe the outer shell and sensor areas, especially the sensor parts such as cameras.

5.7.2 Inspection and Maintenance

1. Visually and manually inspect whether the Robot body, various joints and connections are intact. If there are cracks or damages, replace the relevant parts in a timely manner.
2. Check whether the drive wheels and universal wheels are entangled with foreign objects.
3. Check whether the screws of various connecting parts are tightened, especially the locking screws of joint connectors.
4. Check whether the heat dissipation vents are blocked by foreign objects.
5. Check whether the cooling fan is working normally and whether there is any rubbing noise.
6. Check whether the battery charging port has foreign objects or deformation.
7. Check whether all rockers of the remote control are in the neutral position and whether there is sand or other foreign objects in the gaps of the rockers.
8. Check whether the buttons of the remote control are stuck.

6. Notes

6.1 Operation Specifications

1. Operators must receive official training from UBTECH, be familiar with the Robot's functions, safety instructions and emergency handling procedures. Unauthorized personnel are prohibited from operating the Robot.
2. When modifying Robot parameters (such as load threshold, movement speed), it is necessary to report and record in advance. After modification, test and verify must be conducted to confirm that there is no problem before putting the Robot into use. Random modification of core parameters is prohibited.
3. Before the Robot performs a new task (such as a part installation task that has not been trained), it must first be tested in a simulated environment, then a real-machine test operation must be conducted. Direct execution of untested tasks is prohibited to avoid equipment failures.
4. The user shall bear full responsibility for any personal injury or property loss caused by non-compliance with the operation specifications.

6.2 Battery Maintenance

1. Avoid overcharging/over-discharging the battery. Power off the battery in a timely manner after use.
2. It is not recommended to keep the charger plugged in for a long time.
3. The battery is a built-in component and cannot be disassembled. Check the battery level once a month to ensure that the battery level is maintained between 50% and 80%.
4. For long-term storage, it is recommended to keep the battery level between 50% and 70% before storing the entire machine. Charge the 46Ah battery once every 2–3 months; regular charging can effectively avoid capacity attenuation caused by deep discharge and extend the battery service life.
5. If the battery ages, swells or the battery life decreases sharply, contact technical support for replacement in a timely manner.

6.3 Storage Recommendations

1. When the Robot is out of use for a long time (more than 1 month), turn off the power and store the Robot in a dry, ventilated and dust-free environment with a temperature of 0~30°C and a humidity of less than 75% RH. Avoid direct sunlight or humid environments.
2. Do not store the Robot with corrosive gases, flammable materials or strong electromagnetic sources, and do not place it in direct sunlight.
3. It is recommended to store the Robot in the original shipping case during storage.

6.4 Fault Handling

1. When the Robot malfunctions, first record the fault phenomena (such as screen prompts, abnormal noise location, fault time), then contact the official UBTECH service team. Unauthorized disassembly or maintenance of core components is prohibited.
2. In case of emergency accidents such as equipment collision or battery fire, trigger the emergency stop switch immediately, organize personnel evacuation, and use a dry powder fire extinguisher to put out the fire (water is prohibited for fire extinguishing).

6.5 Intellectual Property Rights

1. The intellectual property rights of the Robot's hardware design, software programs, control algorithms, etc. belong to UBTECH. Unauthorized copying, modification or dissemination of relevant technical materials is prohibited. Violators shall bear legal responsibilities and compensate UBTECH for any losses caused thereby.
2. Reverse engineering or cracking of the Robot is prohibited to prevent the leakage of core technologies. If any infringement is found, UBTECH will pursue legal responsibilities, and the user shall compensate UBTECH for any losses caused thereby.

6.6 Legal and Compliant Use of Large Models

The Cyberspace Administration of China, together with the National Development and Reform Commission, the Ministry of Education, the Ministry of Science and

Technology, the Ministry of Industry and Information Technology, the Ministry of Public Security, and the National Radio and Television Administration, has issued the Interim Measures for the Administration of Generative Artificial Intelligence Services (hereinafter referred to as the "Measures"), which came into force on August 15, 2023. Customers shall comply with the relevant requirements of the Measures when using the large model-related functions of our Robot.

7. Version Revision History

Version	Date	Modification Content
V1.0	2025/8/24	Initial Version